

PantryPilot

Final Project Presentation

Smart meal and grocery planning with safety-aware software quality controls

System

Meal plans, substitutions, pantry use, and grocery lists

Quality focus

Risk, CM, traceability, and system-level testing

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Assignment coverage map

- 1 ———— **Project introduction and system features**
- 2 ———— **Role of each group member**
- 3 ———— **Risk identification and tracking**
- 4 ———— **Configuration management approach**
- 5 ———— **System testing approach and outcome**

Central message

PantryPilot combines useful automation with disciplined SQA: high-risk behavior is identified, controlled, versioned, and tested.

Project introduction: why PantryPilot exists

Planning friction

Users often decide meals, check ingredients, and build grocery lists in separate tools.

Constraint complexity

Allergies, disliked ingredients, budgets, nutrition goals, and preferences can conflict.

Food waste

Pantry items expire while shopping decisions ignore what is already available at home.

PantryPilot answer

A smart meal and grocery planning system that maintains profiles, recipes, pantry inventory, weekly meal plans, AI substitutions, and consolidated grocery lists.

System features and release scope

V1.0 baseline

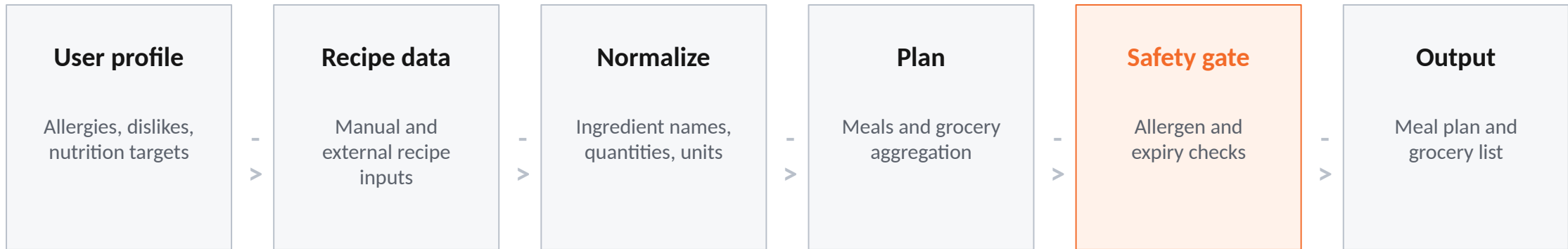
- Profile and dietary constraints
- Recipe library and normalized ingredients
- Weekly meal planning
- Consolidated grocery aggregation
- Basic pantry-aware planning model

V1.1 safety and grocery controls

- Deterministic allergen conflict detection
- Safety gate before AI output reaches the user
- Expiry-window logic for pantry items
- Grocery unit aggregation regression coverage
- CI workflow for repeatable checks

Future V2 scope: nutrition optimizer, pantry-aware ranking, and generated recipes after the safety baseline is stable.

Software design: safety-aware planning pipeline



Design decision

Use AI only for suggestion value; keep safety decisions deterministic, reviewable, and testable.

Role of each group member

This was completed as an individual project, so Zixuan Liang owned each presentation section and project artifact.

Project and requirements

Defined the PantryPilot concept, V1/V1.1 scope, requirements, adverse conditions, and quality goals.

Risk management

Built and updated the risk table, likelihood-impact scores, mitigation strategy, and weekly tracking history.

Configuration management

Created the generic Git repository package, branch model, release tags, changelog, CI workflow, and CM report.

System testing

Designed six system-level tests, executed the runner, captured results, and documented environment/setup details.

Risk identification and tracking method

Identification

Risks were identified from safety-sensitive outputs, external dependencies, optimization complexity, data handling, and release scope.

Scoring

Each risk received a probability score and an impact score from 1 to 5; total priority equals probability times impact.

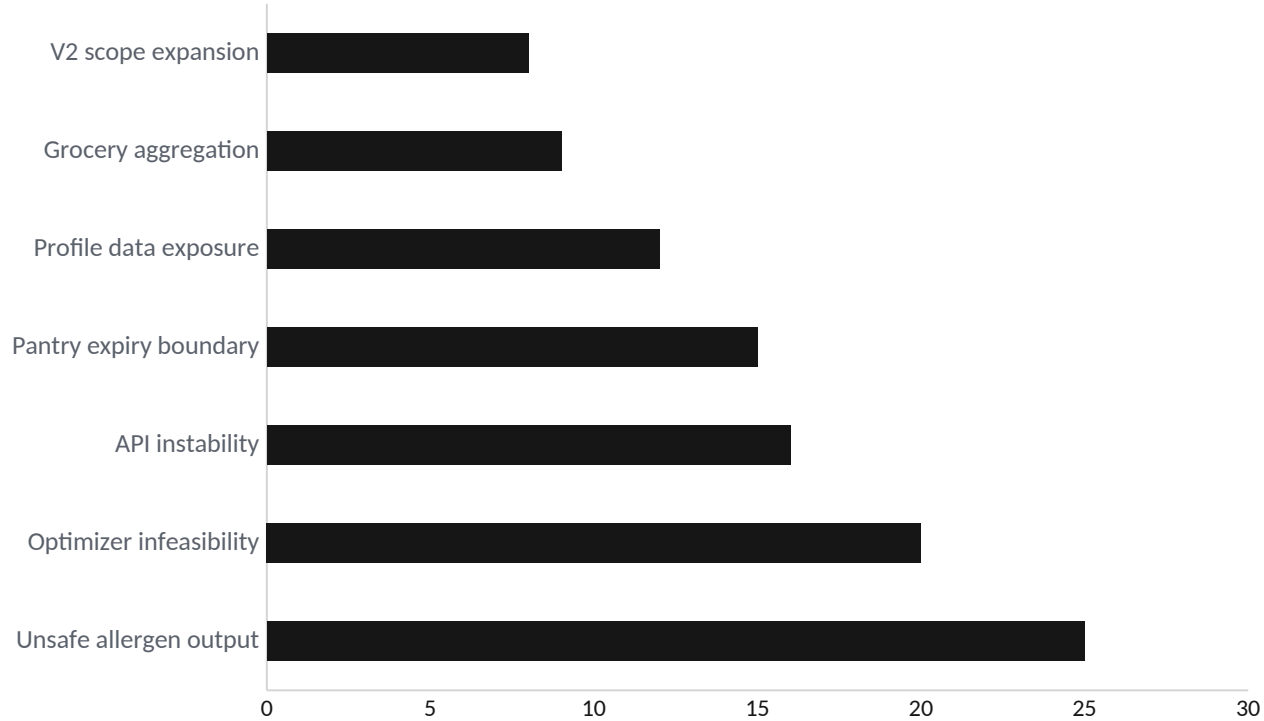
Tracking

Risk status was reviewed weekly and linked to controls, tests, CI checks, release notes, or retirement rationale.

Current risk posture

R1-R5 remain active and controlled; R6 is monitored; R7 scope expansion was reduced and retired after the V1.1 boundary was clarified.

Top project risks by score



Priority conclusion

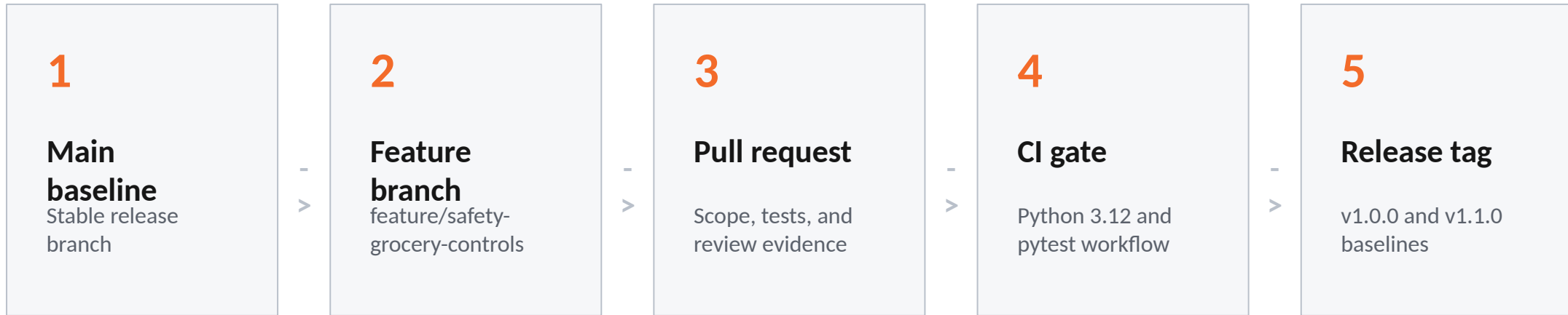
- Highest risk: unsafe LLM output reaching an allergic user
- Critical technical risk: optimizer infeasibility or runtime failure
- High operational risks: unstable external recipe data and pantry date boundaries
- Privacy risk remains active because profiles contain sensitive preferences

How the main risks were reduced

Risk	Reduction control	Verification evidence
Unsafe allergen output	Deterministic allergen filter blocks conflict before display	System test V1.1-ST-01 returned safe=false for soy conflict
Optimizer failure	Bounded solver concept and explicit future relaxation order	Kept out of V1.1 release baseline
Recipe API instability	Offline fixtures/cache strategy and normalized ingredient model	System tests avoid external network dependency
Pantry expiry boundary	Inclusive 3-day rule; expired items excluded	V1.1-ST-03 passed today/+3/+4/yesterday boundary checks
Grocery aggregation	Canonical unit aggregation by ingredient name and unit	V1-ST-03 produced rice quantity 3.0 cup

Mitigation rule: a major risk is not considered controlled until it has a design control and testable evidence.

Configuration management approach



Controlled items

Source code, tests, configuration files, CI workflow, changelog, release notes, risk table, CM report, and system test report.

Configuration management evidence

Evidence	Project record
Repository package	PantryPilot_CM_Generic_Files
Branches	main; feature/safety-grocery-controls
Release tags	v1.0.0 baseline; v1.1.0 safety and grocery controls
Key commits	842ae5e baseline; 72a997e feature controls; 834e64e merge/release
CI workflow	.github/workflows/ci.yml installs requirements and runs pytest on Python 3.12
Traceability	CHANGELOG, release notes, tests, and CM report preserve change history

System testing approach

Methodology

Manual/system-level scenarios exercise release identity, quality gates, grocery aggregation, safety controls, and pantry date boundaries.

Environment

macOS 26.5.1 arm64, local Python 3.11.15, CI Python 3.12, pytest 8.2.2 in requirements.

Setup

Clone/source package, install requirements, run pytest in CI, and run python3 system_test_runner.py for system scenarios.

Test class	Coverage target
Release readiness	Version identity and required quality gates
Core workflow	Grocery aggregation across recipes and normalized units
Safety behavior	Allergen conflict blocked; safe recipe allowed
Boundary conditions	Pantry item expires today, +3 days, +4 days, and yesterday

System testing outcome

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system scenarios passed

No defects were observed during the final local system-test execution.

Scenario	Actual result
V1-ST-01 release identity	PantryPilot 1.1.0: Safety & Grocery Controls
V1-ST-02 quality gates	ci_required, pull_request_required, review_required, unit_tests_required
V1-ST-03 grocery aggregation	rice quantity 3.0 cup
V1.1-ST-01 allergen conflict	soy conflict detected; safe=false
V1.1-ST-02 safe recipe	safe=true
V1.1-ST-03 expiry window	today=true, +3=true, +4=false, yesterday=false

Final takeaway and references

PantryPilot's quality story is not just adding intelligence to recipes.

It is a controlled system where risky outputs are bounded by deterministic checks, versioned baselines, risk tracking, and system-level test evidence.

Project sources

- PantryPilot Risk Management Report
- PantryPilot Final Configuration Management Report
- PantryPilot System Testing Report
- PantryPilot CM Generic Files source package

Questions